

RaktaScan

AI-Powered Non-Invasive Anemia Screening

Problem Statement Omni_BioTech_2 — Non-Invasive Anemia Screening	Domain BioTech & HealthTech
Team Name Duo Tech	Team Members Vetha Narayanan G Akshaya I

1. Problem Understanding

Anemia is one of India's largest and most under-addressed public health problems. According to the National Family Health Survey (NFHS-5, 2019-21), **57% of women aged 15-49** and **67% of children aged 6-59 months** in India are anemic — among the highest burdens of any country, and the prevalence has risen since the previous survey round rather than fallen.

The core barrier is not awareness, it is access to screening. Diagnosing anemia today relies on a blood draw (finger-prick or venous sample) processed with lab equipment or reagent strips, which requires trained personnel, consumables, and turnaround time. This makes frequent, mass-scale screening — through ASHA/ANM community health visits, school health checkups, or antenatal checkups — logistically expensive and hard to sustain in resource-constrained and rural settings. As a result, anemia in women and children very often goes undetected until it is symptomatic, delaying the nutritional or medical intervention that could have prevented it.

THE CORE GAP

A low-cost, equipment-light, smartphone-based first-line screening tool that can flag who may need a confirmatory blood test — without pretending to replace one.

2. Proposed Solution

RaktaScan is a camera-based, AI-powered screening web app. The user (or a community health worker) photographs the inner eyelid (palpebral conjunctiva) — a highly vascular, melanin-unaffected tissue whose pallor is a well-studied proxy for hemoglobin level — using a **commonly available smartphone camera under guided capture conditions**. An on-screen alignment overlay helps maintain consistent framing and lighting.

The app automatically detects and crops the conjunctiva region, validates image quality, analyzes pallor using a lightweight trained deep learning model, and returns an instant risk tier — **Low / Moderate / High risk of anemia** — along with a clear recommendation to get a confirmatory blood test at a nearby health center for Moderate/High results.

A Community Health Worker Mode lets one person screen many people in a single session (e.g. an anganwadi or school visit) and export a summary for follow-up. The app is designed offline-first, so it keeps working in low-connectivity areas.

RaktaScan is explicitly positioned as a screening / triage aid, not a diagnostic device — its job is to nudge the right people toward a real blood test sooner, not to replace one.

3. What Makes RaktaScan Different

Innovation	How it improves real-world screening
AI-Guided Capture	Real-time framing guidance helps users capture the relevant conjunctiva region consistently.
Image Quality Gate	Checks blur, lighting, alignment, ROI visibility and capture quality before inference; poor images trigger recapture instead of an unreliable prediction.
Automatic ROI Detection	Locates and crops the conjunctiva region for standardized preprocessing.
Lightweight On-Device AI	MobileNetV3 is selected for low-latency inference and offline-first operation.
Explainable Risk Assessment	Provides Low / Moderate / High screening risk with plain-language guidance rather than only a raw probability.
Community Health Worker Mode	Supports batch screening, session summaries and follow-up workflows.
Offline-First + Privacy	Prioritizes on-device processing and minimizes unnecessary retention of sensitive health images.

4. System Architecture

1. Smartphone Camera User / CHW capture	2. Guided Capture Alignment + lighting guidance	3. Image Quality Gate Blur / light / alignment / ROI
4. Conjunctiva ROI Detection Locate and crop relevant region	5. Preprocessing Normalize image for inference	6. MobileNetV3 Transfer-learned screening model
7. Risk Assessment Low / Moderate / High	8. Explainable Result Plain-language screening guidance	9. Confirmatory Test Recommendation Moderate/High → clinical confirmation

Supporting components: Community Health Worker Dashboard + secure screening history. Optional backend supports heavier processing and CHW batch mode.

5. Key Features

- Guided photo capture with a real-time alignment overlay for consistent, reliable image quality
- AI Image Quality Gate that rejects unsuitable images before inference
- Automatic conjunctiva ROI detection and preprocessing
- One-tap AI risk analysis — Low / Moderate / High, with a plain-language explanation
- Clear referral guidance for Moderate/High risk results, pointing to the nearest testing/health facility
- Community Health Worker Mode for batch-screening multiple people in one session, with exportable summaries
- Screening history & trend tracking — useful for repeat monitoring, e.g. across a pregnancy
- Multilingual UI (Hindi + regional language toggle) for rural and low-literacy accessibility
- Offline-capable, on-device inference so it works in low-connectivity areas without a data connection
- Privacy-preserving image handling with encrypted storage when screening history is retained

6. Technology Stack

Layer	Technology
Frontend	React.js (Vite) as a responsive PWA, Tailwind CSS
Camera / landmark detection	MediaPipe Face Landmarker (JS) — auto-locates the eye region for guided capture
Primary Model	MobileNetV3 with transfer learning; selected for lightweight deployment. Lighting/rotation augmentation will be used during training.
Alternative Model	EfficientNet-B3 — evaluated as an accuracy-oriented alternative during experimentation
On-device inference	ONNX Runtime Web for offline-first browser inference
Backend	FastAPI (Python) — model serving for CHW batch mode and heavier processing
Database	Supabase / PostgreSQL — screening history for CHW mode
Hosting	Vercel (frontend), Render / Hugging Face Spaces (backend & model)
Version control	GitHub

7. Implementation Plan

Phase	Window	Milestones
Phase 1 — Idea Submission	15–20 Aug 2026	Finalize architecture; source/curate reference conjunctiva dataset; build capture-UI wireframes; submit this idea PDF.
Phase 2 — Progress Submission	21–27 Aug 2026	Build camera capture + automatic ROI detection + Image Quality Gate; train baseline MobileNetV3; evaluate using precision, recall, F1-score, ROC-AUC and confusion matrix ; wire up capture-to-prediction-to-result flow; prepare screenshots/demo for PPT.
Final Submission	28 Aug – 1 Sep	Add CHW mode, history tracking and multilingual UI; refine model and robustness; polish UI/UX; deploy live app; finalize GitHub repo, README and demo video.

8. Success Metrics

AI: Precision, Recall, F1-score, ROC-AUC and confusion matrix on held-out evaluation data.

System: Image-quality validation, prediction latency and offline inference capability.

UX: Clear capture guidance, understandable risk communication and accessibility across supported languages.

Important: No performance numbers will be claimed before the model is actually evaluated.

9. Scalability & Impact

Individual	→	Family	→	School	→	Anganwadi
Community Health Worker	→	Public Health Programs				

10. Responsible AI & Safety

RaktaScan is a screening / triage aid, NOT a diagnostic device.

Moderate/High screening risk results recommend confirmatory hemoglobin testing. The system will not make unsupported medical claims. Image-quality failures trigger recapture, and privacy-preserving/on-device processing is prioritized; retained screening history is encrypted.

References

- Ministry of Health & Family Welfare, Government of India — National Family Health Survey (NFHS-5), 2019-21.
- A Machine Learning Method to Screen Anemia From Conjunctiva Images Taken by Smartphone, IEEE, 2023.
- Detection of Anemia Using Conjunctiva Images: A Smartphone Application Approach, ScienceDirect, 2023.
- Deep Learning Model-Based Detection of Anemia from Conjunctiva Images, Healthcare Informatics Research, 2025.